Turmeric Drying Kinetics and Volumetric Shrinkage Dataset

Experimental convective drying data at 50°C, 60°C, and 70°C | Sources: Gagare et al. (2017); Parveen et al. (2013)

INITIAL MOISTURE CONTENT	INITIAL VOLUME	MAX VOLUMETRIC SHRINKAGE	TEMPERATURE CONDITIONS
411.25 % (db)	32.20 cm³	76.80%	50°C, 60°C, 70°C

Temperature (°C)	Drying Time (h)	Volume (cm³)	Volumetric Shrinkage (%)	Moisture Ratio (MR)	Moisture Content (db, %)
50°C	0	32.20	0.00%	1.00	411.25
50°C	2	28.15	12.58%	0.84	345.50
50°C	4	24.60	23.60%	0.70	288.10
50°C	6	21.45	33.39%	0.58	237.20
50°C	8	18.72	41.86%	0.47	192.80
50°C	10	16.36	49.19%	0.38	154.60
50°C	12	14.38	55.34%	0.30	122.40
50°C	16	11.48	64.35%	0.18	75.30
50°C	20	9.60	70.19%	0.11	44.80
50°C	24	8.52	73.54%	0.06	26.50
50°C	29	7.92	75.40%	0.04	15.40
50°C	36	7.61	76.37%	0.02	10.15
50°C	48	7.47	76.80%	0.02	7.81
60°C	0	32.20	0.00%	1.00	411.25
60°C	2	26.10	18.94%	0.76	312.40
60°C	4	21.30	33.85%	0.57	234.60
60°C	6	17.45	45.81%	0.42	172.50
60°C	8	14.50	54.97%	0.30	124.30

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Temperature (°C)	Drying Time (h)	Volume (cm³)	Volumetric Shrinkage (%)	Moisture Ratio (MR)	Moisture Content (db, %)
60°C	10	12.21	62.08%	0.21	87.20
60°C	12	10.52	67.33%	0.15	59.80
60°C	16	8.65	73.14%	0.07	28.60
60°C	20	7.92	75.40%	0.04	15.20
60°C	24	7.62	76.34%	0.03	10.40
60°C	29	7.51	76.68%	0.02	8.45
60°C	36	7.47	76.80%	0.02	7.81
70°C	0	32.20	0.00%	1.00	411.25
70°C	2	23.82	26.02%	0.67	275.60
70°C	4	18.06	43.91%	0.44	182.40
70°C	6	14.14	56.09%	0.29	118.50
70°C	8	11.41	64.57%	0.18	74.20
70°C	10	9.63	70.09%	0.11	45.30
70°C	12	8.54	73.48%	0.07	26.80
70°C	14	7.99	75.19%	0.04	16.70
70°C	16	7.70	76.09%	0.03	11.80
70°C	20	7.54	76.58%	0.02	9.10
70°C	24	7.48	76.77%	0.02	8.05

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Temperature (°C)	Drying Time (h)	Volume (cm³)	Volumetric Shrinkage (%)	Moisture Ratio (MR)	Moisture Content (db, %)
70°C	29	7.47	76.80%	0.02	7.81